Chapter 19 Homework Answer Sheet

Name:_____ Student ID:_____ Instructor:_____ Score:_____

Part A multiple choice (18 points)

1-5AAAAA

6-10BBBBB

11-15CCCCC

16-18DDD

Part B short question （20 points）

19 (4 points)

(**a**)(2 points) At P=1 atm, T_m= 0 ℃ and ice cube and surroundings has infinitesimally small difference in temperature.

(**b**)(2 points) No. Melting is an endothermic process that increases molecules distance (entropy of the system). Even though there is no temperature change.

20 (6 points)

(**a**) (2 points)

$\Delta H° = \Delta H_f°[CH_3OH(g)]-1/2\Delta H_f°[O_2(g)]-\Delta H_f°[CH_4(g)]$

$=-201.2-1/2*0-(-74.8)=-126.4$ kJ

$\Delta S° = S°[CH_3OH(g)]-1/2S°[O_2(g)]-S°[CH_4(g)]$

$=237.6-1/2*205-186.3=-51.2$ J/K

(**b**) (1 point)

Increase.

(**c**) (1 point)

$\Delta G° = \Delta H°-T\Delta S°=-126.4-298*(-51.2)/1000=-111.1$ kJ

OR$\Delta G° = \Delta G_f°[CH_3OH(g)]-1/2\Delta G_f°[O_2(g)]-\Delta G_f°[CH_4(g)]$

$=-161.9-1/2*0-(-50.8)=-111.1$ kJ

(**d**) (2 points)

$\Delta G=\Delta H-T\Delta S=-126.4-T*(-51.2)/1000$

T=2468.75 K

No. This temperature is so high that the reactants and products are likely to decompose. At standard conditions, equilibrium is functionally unattainable for this reaction.

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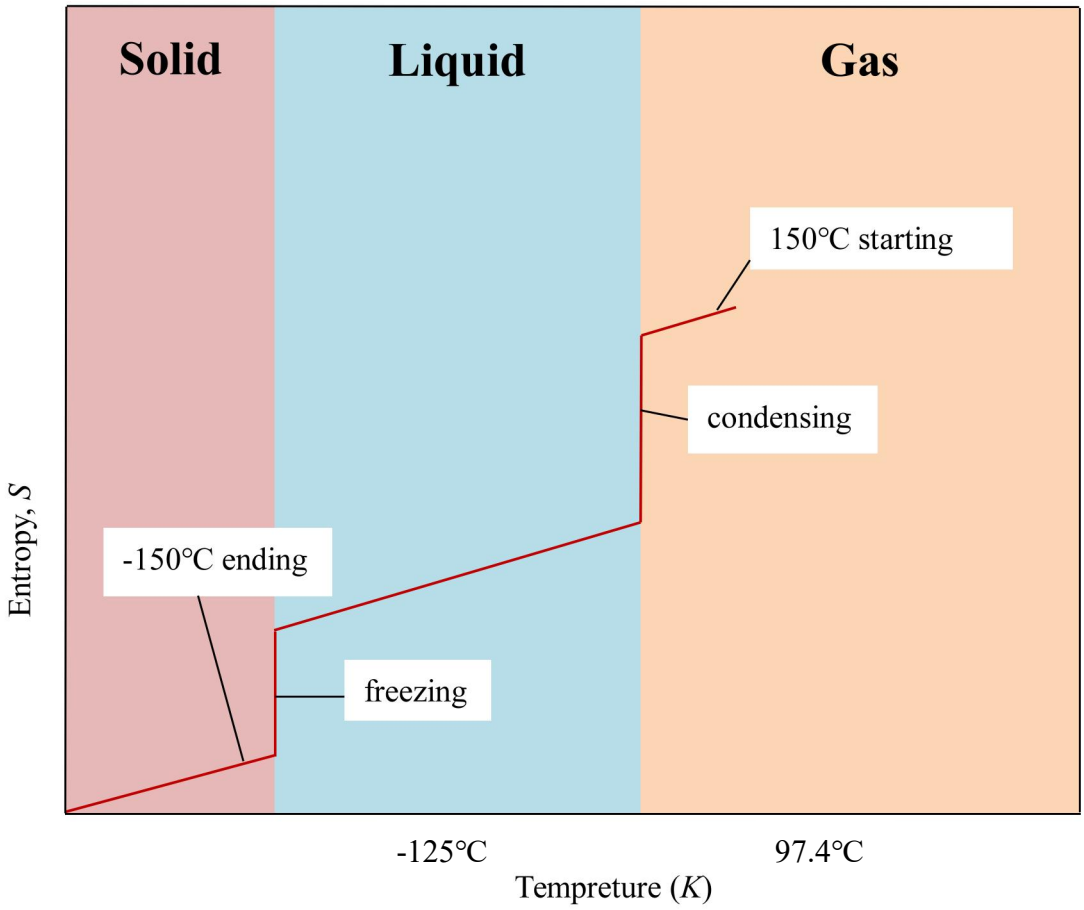
21 (4 points)



22 (6 points)

(**a**) (2 points)

$Br_2(l) \rightleftharpoons Br_2(g)$ P=1atm

(**b**) (2 points)

at equilibrium $\Delta G°=0$

(**c**) (2 points)

$\Delta H°$=30.71 kJ

$\Delta S°$=245.3-152.3=93.0 J/K

$\Delta G° = \Delta H° - T_b \Delta S° = 0$

$T_b = \Delta H°/\Delta S° - 273$ =57.2°C OR 330. K